

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: Numerical Method

Code No: CSC212 Semester: 3rd Semester

Full Marks: 60 Pass Marks: 24 Time: interval, estimate the velocity and acceleration at $x=4$. Time (x) 124810 Distance (y) 01521277. What is application of numerical integration? Find the value of integration for $\int_1^2 e^{1/x} dx$ using Simpson's 3/8 rule with $n=6$. 8. Solve the following system of linear equations using Gauss-Jordan elimination method. $x+2y-3z=42$, $x+4y-6z=8$, $x-2y+5z=49$. Given the data points below X 1.03.04.0f(x) 1.54.59.0 Find cubic spline which belongs to $1 \leq x \leq 3$ and estimate $f(2)$ using cubic splines. 10. What is differential equation? Differentiate between ODE and PDE with example. 11. Solve $dy/dx = x/y$, $y(0) = 1$, at $x=0.4$ using Runge-Kutta's 4th order method. 12. Solve the Poisson equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 64xy$, $0 \leq x \leq 1$, $0 \leq y \leq 1$ with boundary conditions: $u(0,y)=0$, $u(x,0)=0$, $u(1,y)=150$, $u(x,1)=150$ and $h=1/3$.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. What are inherent errors? Derive the Newton Raphson method for solving non-linear equation and using this method solve $x^2-5x+6=0$, calculate upto 3 decimal places.
2. What are the limitations of direct methods for solving a system of linear equations? How Gauss Seidel method differs from Jacobi iteration? Solve the following system of linear equation using Jacobi iteration method. $2x-7y-10z=175$, $x+y+3z=14$, $x+10y+9z=7$
3. Write an algorithm and program to implement Lagrange interpolation method.

Section B

Attempt any Eight questions [8×5=40]

4. Consider the following data points estimate the $f(0.6)$ using Newton's interpolation formula. x 0.10.20.30.40.5f(x) 2.683.043.383.693.97
5. What is regression analysis? Fit a second order polynomial for the following data values. x 246810y 1.42.02.42.62.8
6. What is numerical differentiation? The table below gives the values of distance travelled by a vehicle at various time interval, estimate the velocity and acceleration at $x=4$. Time (x) 124810 Distance (y) 0152127
7. What is application of numerical integration? Find the value of integration for $\int_1^2 e^{1/x} dx$ using Simpson's 3/8 rule with $n=6$.
8. Solve the following system of linear equations using Gauss-Jordan elimination method. $x+2y-3z=42$, $x+4y-6z=8$, $x-2y+5z=4$

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9. Given the data points below
- | x | f(x) |
|-----|------|
| 1.0 | 3.0 |
| 3.0 | 4.0 |
| 4.0 | 1.5 |
| 5.0 | 4.0 |
| 9.0 | 0.0 |
- Find cubic spline which belongs to $1 \leq x \leq 3$ and estimate $f(2)$ using cubic splines.
10. What is differential equation? Differentiate between ODE and PDE with example.
11. Solve $dy/dx = x/y$, $y(0) = 1$, at $x=0.4$ using Runge-Kutta's 4th order method.
12. Solve the Poisson equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 64xy$, $0 \leq x \leq 1$, $0 \leq y \leq 1$ with boundary conditions: $u(0,y)=0$, $u(x,0)=0$, $u(1,y)=150$, $u(x,1)=150$ and $h=1/3$.